

Model classification

Modelling Fall 2025

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Grouping by opposite pairs

- linear versus nonlinear;
- steady state versus non-steady state;
- lumped parameter versus distributed parameter;
- continuous versus discrete variables;
- deterministic versus stochastic;
- interpolation versus extrapolation;
- mechanistic versus empirical;
- coupled versus not coupled.

importance ?

How can we solve it?

How does the output look?

What assumptions can be made?

How much can we trust it?

Can we use it to understand underlying mechanisms?

Examples of model descriptions in literature

free concentration. This extrapolation model considers the different composition of human serum and cell culture media concerning lipid content and concentration of

This paper provides a review of computationally efficient approaches to nonlinear model predictive control. The methods considered cover the following areas: tailoring of

Empirical models often prove inadequate in that they are limited by the range of data on which

A discrete stochastic model of the COVID-19 outbreak: Forecast and control
data compilation to calibrate a mechanistic model of ecosystem dynamics based on grazing

In this article, lumped parameters of quick coupling is obtained by using Solid Works Flow Simulation

Linear vs nonlinear model

The **dependent variable** must:

1. appear in the first power (or zero)
2. Not be in a nonlinear function
3. Not be multiplied/divided with other dependent variables

$$y = 2x + 5$$

$$y = x^3 - x^2$$

$$\frac{dy}{dx} = \sin(xy)$$

$$\frac{dy}{dx} = yx^2 - y$$

$$\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2 + y$$

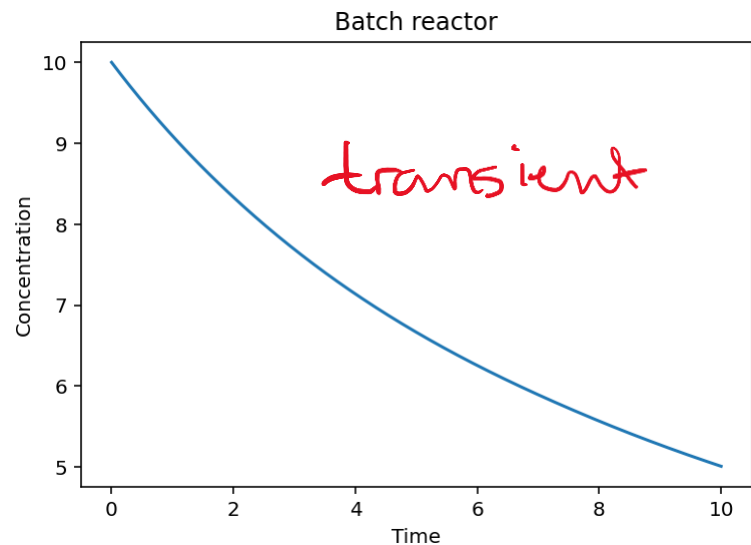
Nonlinear

Steady-state versus transient

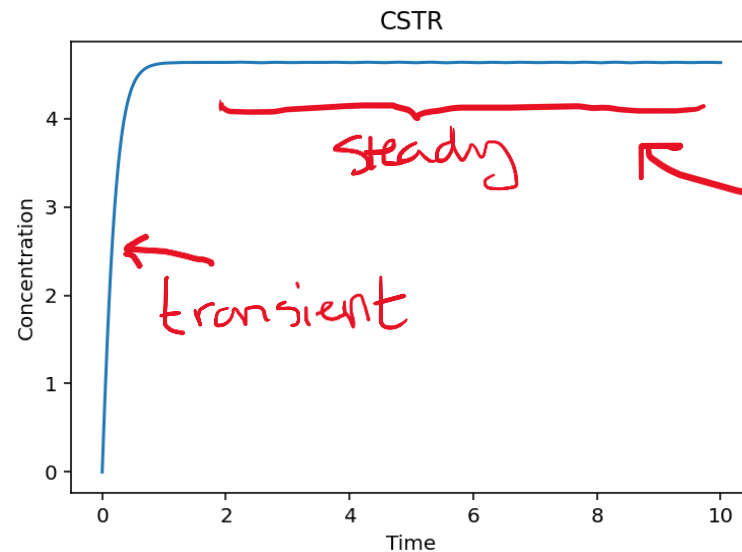
Transient/dynamic/non-steady-state

The dependent variable changes with time

Batch reactor (first order reaction) $\frac{dC}{dt} = -kC$



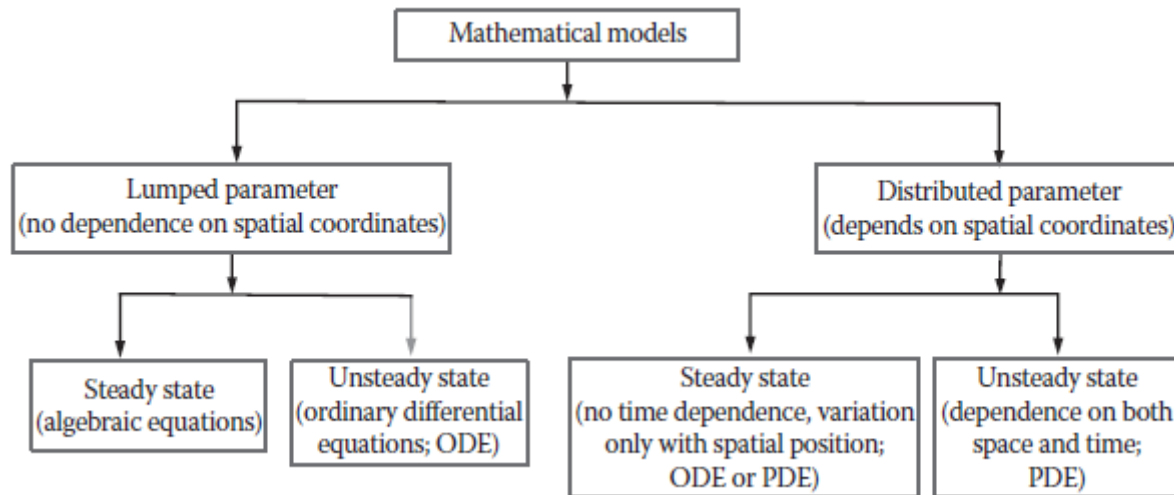
CSTR (first order reaction) $\frac{dC}{dt} = \frac{Q}{V}(C_{in} - C) - kC$



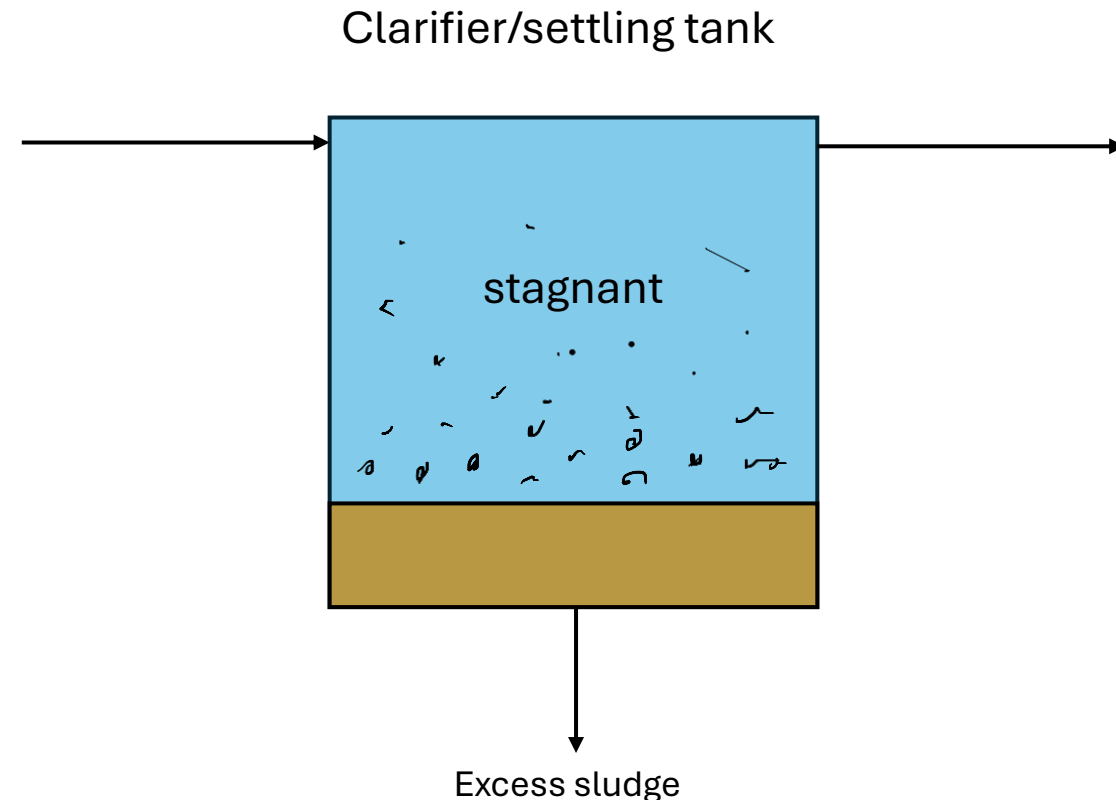
Lumped parameter versus distributed parameter

Does state variable and parameters vary in space? Closely related to PDEs!

Models are often hybrids with some parameters being lumped and some being distributed



Binay Kanti Dutta. 2020. Mathematical Methods in Chemical and Biological Engineering

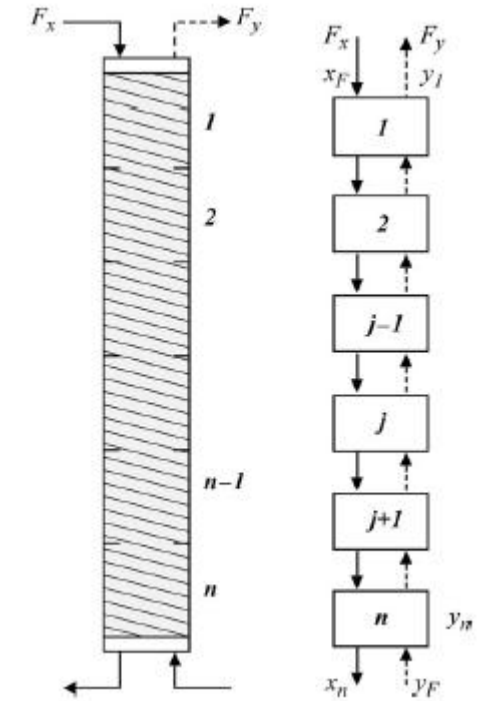


Continuous and discrete models

differential equation: $\frac{dy}{dt} = -kC$

difference equation: $y_{i+1} = y_i + \Delta t \cdot f(t_i, y_i)$

Looks familiar?



Algebraic equation

Algebraic equations contain no derivatives (involves variables and constants only)

$$y = 2x + 5$$

$$y = x^3 - x^2$$

The finite difference method converted to system of algebraic equations

An equation or system with both differential and algebraic equations is a "Differential-algebraic equation/model"

$$\begin{bmatrix} 1 & 0 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & & 1 \end{bmatrix} \begin{bmatrix} y_0 \\ y_1 \\ \dots \\ y_{n-1} \\ y_n \end{bmatrix} = \begin{bmatrix} 0 \\ -gh^2 \\ \dots \\ -gh^2 \\ 50 \end{bmatrix}$$

Mechanistic or empirical

Mechanistic: based on underlying laws of physics and chemistry (ideally we do not need data to formulate the model).

$$\frac{d^2h}{dt^2} = g \quad \text{position of falling object in vacuum (mechanistic)}$$

$$\frac{dX}{dt} = \frac{u_{max} \cdot S}{K_S + S} \cdot X \quad \text{growth of bacteria in a batch reactor (semi-mechanistic)}$$

$$y = A \cdot x + B \cdot x + C \cdot x^2 \quad \dots \text{Enteric methane produced by a dairy cow, where } x \text{ is feed intake, } A$$

is age of cow, B is % of fat in diet, C is number of hours spend grazing in the field. (finding correlations, but not necessary causality)

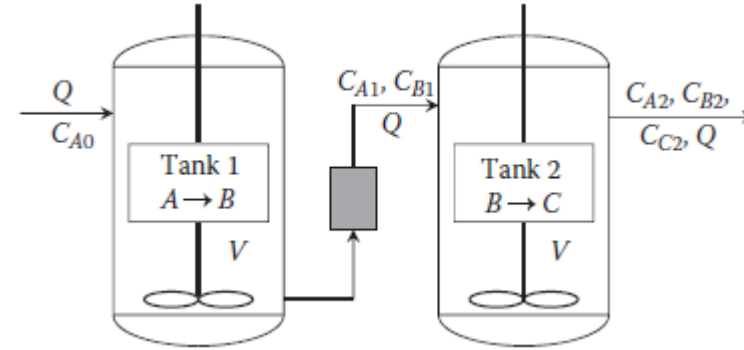
Coupled models

Mass balance on B in two reactors

$$\frac{dC_{B1}}{dt} = k_1 C_{A1}^2 - \frac{Q}{V_1} \cdot C_{B1}$$

$$\frac{dC_{B2}}{dt} = -k_2 C_{B2}^2 - \frac{Q}{V_2} \cdot C_{B2} + \frac{Q}{V_2} \cdot C_{B1}$$

Weak coupling

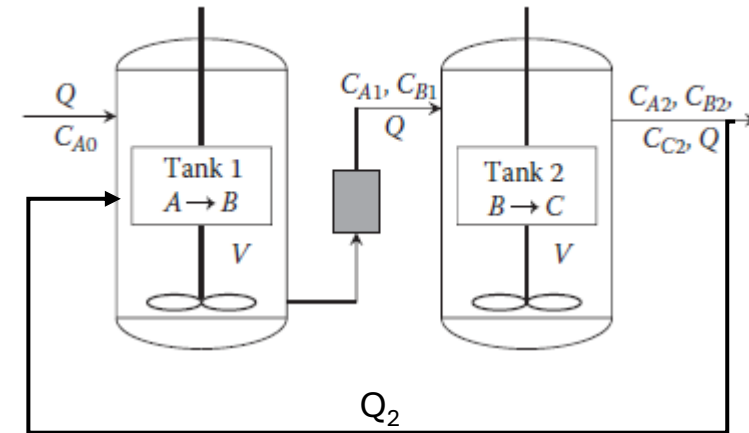


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$$\frac{dC_{B1}}{dt} = k_1 C_{A1}^2 - \frac{Q}{V_1} \cdot C_{B1} + \frac{Q_2}{V_1} \cdot C_{B2}$$

$$\frac{dC_{B2}}{dt} = -k_2 C_{B2}^2 - \frac{Q + Q_2}{V_2} \cdot C_{B2} + \frac{Q + Q_2}{V_2} \cdot C_{B1}$$

Strong coupling



Deterministic or stoichastic

Deterministic models have fixed numbers for parameters
(inherently there is no uncertainty)

Example: Income = hourly rate * hours worked

Stoichastic model are somewhat random

Example: Chance of rolling 6

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$P(X = i) = \frac{1}{6}, \quad i \in S$$

Example2: Chance of heart attack?

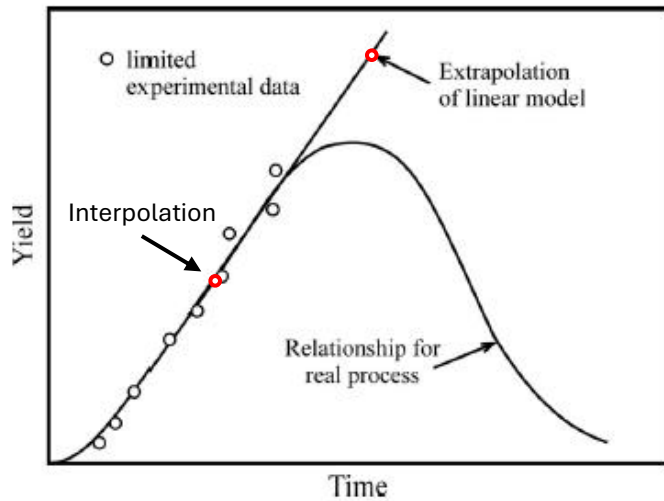
$$P(\text{heart attack in next 15 min}) = 1 - e^{-\lambda \cdot 15}$$

Deterministic universe?

Interpolation/extrapolation

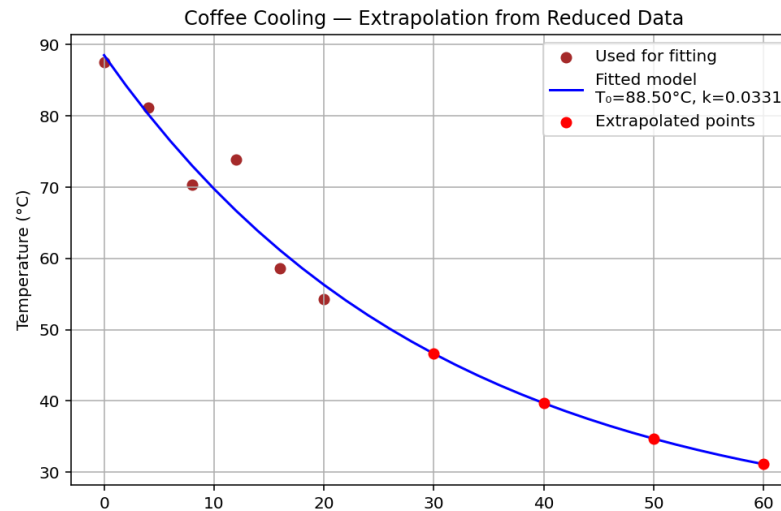
Applying a model within observed data range is interpolation

Applying it outside of the observed data range is risky and is called extrapolation

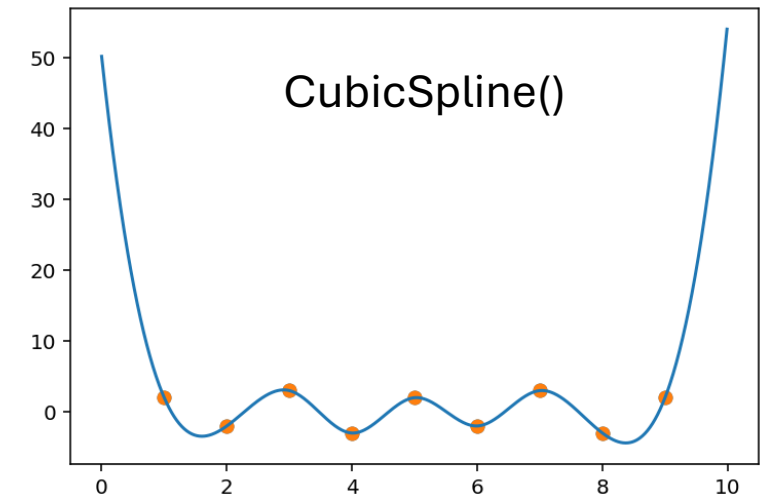


Depending on the model it is more or less risky to extrapolate

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$$



$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3 \text{ for } x_i \leq x \leq x_{i+1}$$



Complexity

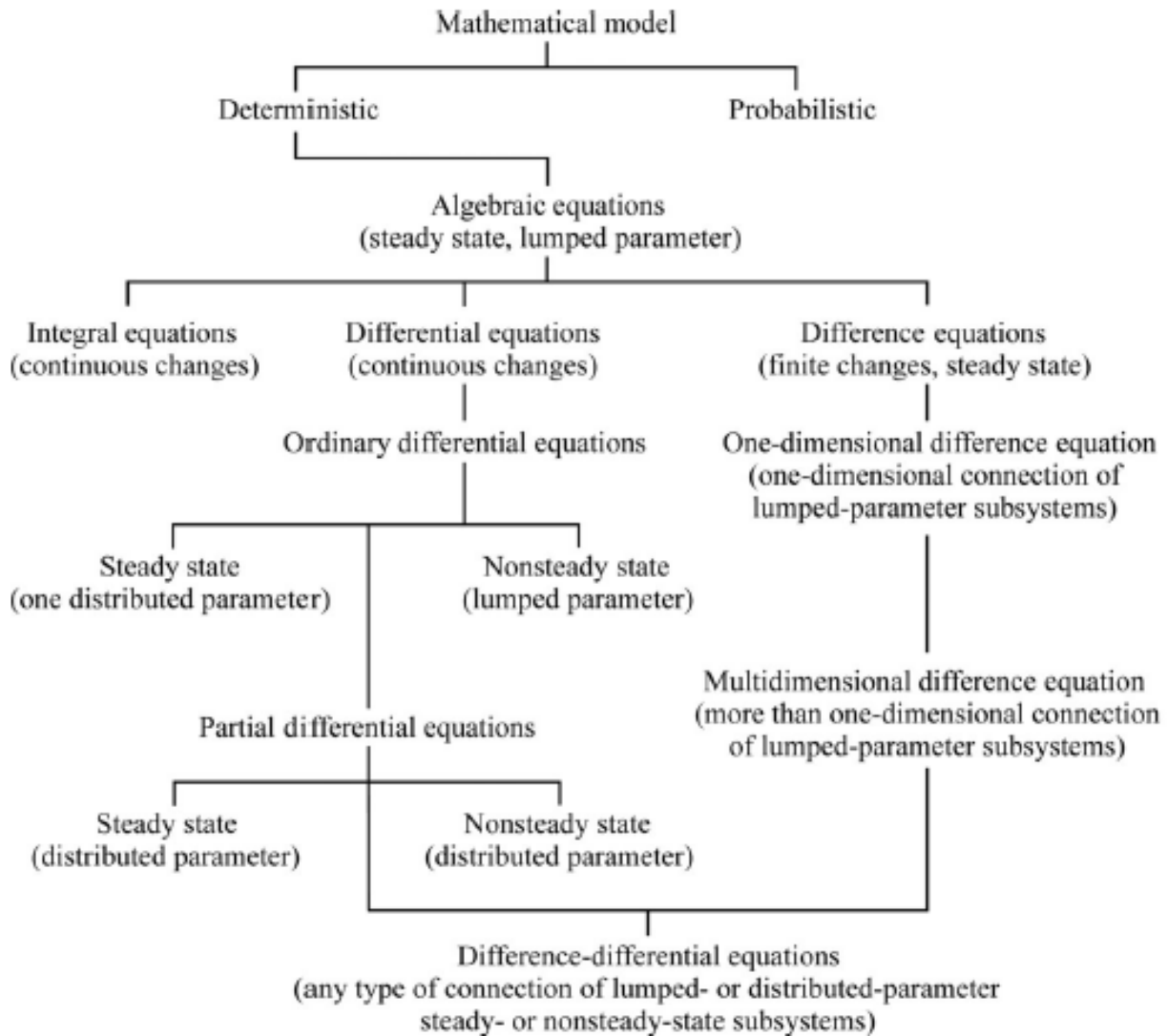


Table 2.1. Classification of mathematical problems and their ease of solution using analytical methods

Equation	Linear equations			Non-linear equations		
	One equation	Several equations	Many equations	One equation	Several equations	Many equations
Algebraic	trivial	easy	essentially impossible	very difficult	very difficult	impossible
Ordinary differential	easy	difficult	essentially impossible	very difficult	impossible	impossible
Partial differential	difficult	essentially impossible	impossible	essentially impossible	impossible	impossible

Model scale

Detail of internal system

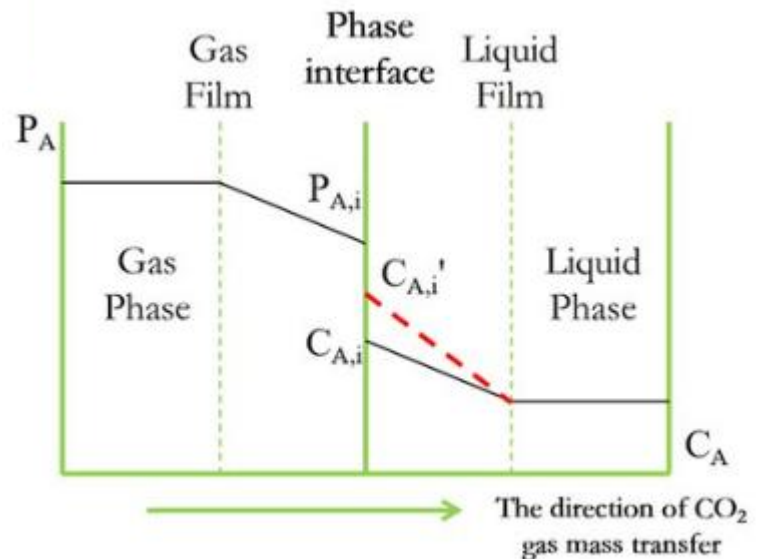
Table 2.2. Classification of models according to scale

Level of physicochemical description	Topical designations	Parameters
Molecular	treats discrete entities; quantum mechanics, statistical mechanics, kinetic theory	distribution functions; collision integrals
Microscopic	laminar transport phenomena, statistical theories of turbulence	phenomenological coefficients; viscosity, thermal conductivity, diffusivity
Mesoscopic	laminar and turbulent transport phenomena; transport in porous media	“effective” transport coefficients
Macroscopic	process engineering, unit operations	interphase transport coefficients

Often involves stoichastic models

Distributed parameter models

Lumped models



$$\frac{\partial C}{\partial t} = D_{CO_2} \frac{\partial^2 C}{\partial z^2}$$

Microscopic approach: descretization, diffusion

$$\frac{d(C_G G)}{dz} = -K_G a (C_G - C_G^*)$$

Macroscopic approach: average bulk concentrations, average mass transfer coefficient

Menti quiz:



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